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Management and coordination of data for digital therapeutics trials

Abstract

Provided herein are systems and methods of managing access to data associated with trials for digital therapeutics on participants. A server can maintain a data storage for a plurality of phases associated with a trial for a digital therapeutic on participants. The data storage may include (i) a first portion of a plurality of datasets from the plurality of participants from one or more of the plurality of phases of the trial, and (ii) a second portion of the plurality of datasets from the digital therapeutics in at least one of the plurality of phases of the trial. The server can receive from a user device, a request identifying a phase of the plurality of phases for which the plurality of datasets is to be accessed. The server can select a corresponding portion of the plurality of datasets based on the request to provide the user device.

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Background/Summary

BACKGROUND

(1) Multiple databases may be used to store and collect data in accordance with a database schema, and may be used to facilitate storage and maintenance of organized data for a clinical trial testing the efficacy and safety of pharmaceuticals. Before a pharmaceutical can be approved by a regulatory agency, the pharmaceutical may undergo multiple clinical trials to test the efficacy and safety of the pharmaceutical, with regulations that, for example, provide privacy to the participants' personal health information (PHI) and personally identifiable information (PII) collected during the trial as well as control participants' enrollment in the trial and how the trial tests the desired endpoints. An endpoint may correspond to the outcome that is being measured by the clinical trial. A pharmaceutical drug, for example, might use survival as an endpoint, comparing the five-year survival rate using an experimental therapy against the five-year survival rate using another treatment or a placebo.

(2) In clinical trials involving pharmaceuticals, such trials may have three phases: (1) the pre-trial phase; (2) the clinical trial phase, which involves administering a pharmaceutical, a medical device, etc.; and (3) the post-trial phase. During the pre-trial and post-trial phases, a clinical trial administrator may approach a participant with a mobile device (e.g., a tablet or laptop), and directly ask questions of the participant regarding the pharmaceutical. The administrator may manually enter the participant responses into a spreadsheet or data entry interface running on the mobile device. In some cases, the clinical trial administrator may communicate with the participant by telephone or video meeting and may alternatively use a desktop device to collect participant responses. The data collected from the participant responses in a trial (and the trial endpoint data) may be held in a clinical trial database called an electronic data capture (EDC).

SUMMARY

(3) Presented herein are systems and methods for managing and coordinating data for digital therapeutics applications. In contrast to trials for pharmaceuticals, the types of data aggregated from trials of digital therapeutics applications may differ greatly from the types of data collected from clinical trials involving pharmaceuticals. A digital therapeutics may correspond to systems and devices implemented in software, hardware, firmware, program logic units, and any combination thereof, that can be used to cure, prevent, mitigate, or treat an array of diseases, disorders, and conditions. The trials for a digital therapeutics may operate differently than clinical trials involving pharmaceuticals in many ways. Notably, during the second phase (e.g., the clinical trial), there may be a participation phase using a digital interface for performing the digital therapeutic activities (alone or in combination with a pharmaceutical). There may be digital lessons or physical activities initiated periodically by the digital therapeutic as well as queries and responses required during the clinical trial. Thus, unique data may be electronically collected during the second phase that does not occur in traditional clinical trials.

(4) To store and maintain the data for digital therapeutics trials, prior approaches may include a clinical database to manage data from the pre-trial phase and post-trial phase, and there is a separate application database to manage data from the trial phase. These two separate databases may not communicate with one another and are maintained by separate entities in separate locations. This may make it difficult to correlate digital therapeutics activity with endpoint data. For instance, a trial endpoint may be on a scale for engagement, and a participant may identify with a low engagement at the pre-trial phase. After performing the clinical trial using the digital therapeutic, however, the participant may identify as high engagement at the post-trial phase. Due to the lack of communication or interfacing among the databases for pre-trial, clinical trial, and post-trial data, it can be burdensome to match the participant with the same profile and conclude

that the post-trial data shows that the digital therapeutic caused the increase in engagement.

(5) Another disadvantage with prior approaches may be that because there are regulations (or other rules) for managing data collected during the three phases of a clinical trial, relying on separate databases requires additional effort to ensure each database complies with the corresponding regulations.

(6) In addition, similar to clinical trials for pharmaceuticals or medical devices, prior approaches may involve a new database built for each new digital therapeutics trial that is evaluating a new digital therapeutic. For example, a new Electronic Data Capture (EDC), new identity verification, new informed consent, and a new entity hosting the EDC may be set up for each new trial. This may involve a substantial amount of time, effort, and cost, each time a new database is built for a new trial.

(7) In such prior approaches, since new databases are created for each new digital therapeutic being evaluated, it may be difficult or impossible to integrate data across the various trials. Thus, more than one digital therapeutics trial may collect information in common, but this data must be duplicated for each database. This may cause unnecessary duplication of data both on the user device and on the servers operating the trials. Unnecessary excess memory and storage usage on the user device is required due to multiple mobile or web apps that the user may be forced to download and operate. Some older mobile devices may not have enough internal memory or storage to operate more than one trial at a time. Inefficiencies may occur because of the inability of data use and sharing among authorized users. Requiring a user to operate more than one mobile or web app can cause confusion because of inconsistent user experience (UX) across multiple mobile or web apps with different user interfaces.

(8) To address these and other challenges, presented herein are systems and methods of managing access to data associated with trials for digital therapeutics on participants. A service may provide a shell to manage the data from all phases of the trial (e.g., pre-trial phase, clinical trial phase, and post-trial phase) in an integrated framework. The framework can be used for multiple digital therapeutics with minimal modification and thus allows for easy and rapid implementation, launch, and testing of a large quantity of multiple new digital therapeutics trials using the same plug-and-play shell. Each new digital therapeutic may be easily plugged in (e.g., plug-and-play) the same framework. One advantage may be the substantial time, effort, and cost that will be saved, from no longer having to build a new infrastructure for each new digital therapeutics trial. For example, the same Electronic Data Capture (EDC), identity verification, informed consent, and entity hosting the EDC can be used for multiple digital therapeutics trials. Regardless of the endpoints for each trial and the procedures involved running the trial, the use of a single framework for housing multiple trials can provide advantages of scale and efficiency.

(9) The service herein may maintain a centralized site (e.g., data lake) where the data from all phases (e.g., pre-trial phase, trial phase, and post-trial phase) of the trial are stored and managed. The advantage in all three phases of the trial being coordinated for a participant in the same clinical trial data storage, may be that there is certainty that the data from a participant referred to in one phase of the trial will correlate to the data from the same participant in a different phase of the trial. For example, when there is data from the pre-trial phase that a participant has low engagement, then data from the post-trial phase that the same participant now has high engagement. Since the data is stored in the same data storage as the data from the trial phase, it can be easily concluded that the digital therapeutic caused the positive increase in engagement.

(10) A centralized data storage makes this correlation much easier to make. Since the participant responses during the trial phase help determine whether the digital therapeutic is the reason for the clinical trial results, it may be useful to maintain the data in a centralized manner. Even when considering multiple digital therapeutics trials, there may be a benefit in that data can be pulled from the same location to coordinate multiple trials involving the same participant.

(11) Furthermore, it may be advantageous to use a common framework, as described herein, on the

user device so that the user (e.g., a clinician) is able to view a single user interface and can launch multiple digital therapeutics trials from one place. For example, multiple digital therapeutics trials can be integrated when their mobile application code and API (application program interface) code is harmonized with pre-defined criteria corresponding to the framework interface. In addition, a user may manage multiple digital therapeutics trials within the framework that are administered or sponsored by different organizations having different site staff. In this case, the site staff may have access only to data for the digital therapeutics trial that they own or coordinate in the clinical trial data storage.

(12) The service herein may manage a closed system in which digital therapeutics user data, the clinical endpoint, and a clinical data manager are all associated with a single user ID and sent to the same centralized data storage. This type of closed system may improve data security and integrity, relative to approaches relying on a multiplicity of different databases. Additionally, the mobile or web app may be executed on a single platform in which multiple digital therapeutics treatments can be delivered, therefore making the solution scalable across many sites and for many digital therapeutics treatments, including third-party treatments.

(13) Aspects of the present disclosure are directed to a method and a system for managing access to data associated with trials for digital therapeutics on subjects. A server can maintain a data storage for a plurality of phases associated with a trial for a digital therapeutic on a plurality of participants. The data storage may include (i) a first portion of a plurality of datasets from the plurality of participants from one or more of the plurality of phases of the trial, and (ii) a second portion of the plurality of datasets from the digital therapeutics in at least one of the plurality of phases of the trial. The server can receive from a user device, a request identifying a phase of the plurality of phases for which the plurality of datasets is to be accessed. The server can select a corresponding portion of the plurality of datasets based on the phase identified in the request. The server can provide the user device access to the corresponding portion of the plurality of datasets in the data storage.

(14) In some embodiments, the server may receive from the user device, a second request to define a plurality of parameters for the trial, the plurality of parameters identifying at least one of: the digital therapeutics, the plurality of phases, and a respective endpoint for each of the plurality of phases. The server may establish the data storage in a data lake in accordance with the plurality of parameters for the trial. In some embodiments, the server may identify a plurality of trials associated with a corresponding plurality of conditions of the plurality of participants provided with the digital therapeutics, and establish a plurality of data storages for the corresponding plurality of trials.

(15) In some embodiments, the server may receive, from the user device, a second request to associate the data storage with a second data storage. The second data storage may include a second plurality of datasets from a second trial separate from the trial associated with the data storage. The server may link, responsive to the second request, the second data storage with the data storage to provide the user device access to the data storage and the second data storage. In some embodiments, the server may identify from a plurality of access privileges, an access privilege for a user associated with the user device, wherein providing the access further includes providing the user device access to the portion of the plurality of datasets, responsive to the access privilege defining that the user is permitted to access the portion.

(16) In some embodiments, wherein receiving the request further includes receiving the request identifying a questionnaire to add to the phase of the plurality of phases, the server may generate, at least one dataset identifying the questionnaire to add to corresponding portion of the plurality of datasets. In some embodiments, wherein maintaining the data storage further includes removing, from the data storage, a first dataset in one portion of the plurality of datasets corresponding to a second dataset in another portion of the plurality of datasets.

(17) In some embodiments, receiving the request further includes receiving the request identifying

a participant of the plurality of participants for which the plurality of datasets is to be accessed, wherein providing the access further includes correlating, within the plurality of datasets on the data storage, one or more datasets associated with the participant identified in the request. In some embodiments, at least one of the plurality of datasets may be generated using at least one of electronic data capture (EDC), identity verification, and informed consent. In some embodiments, at least one of the plurality of participants may be administered with a pharmaceutical to address a condition, at least partially concurrently with use of the digital therapeutics.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The foregoing and other objects, aspects, features, and advantages of the disclosure will become more apparent and better understood by referring to the following description taken in conjunction with the accompanying drawings, in which:

(2) FIG. 1 depicts a block diagram of a system for managing and coordinating data in accordance with an illustrative embodiment;

(3) FIG. 2 depicts a block diagram of a system for managing and coordinating data in accordance with an illustrative embodiment;

(4) FIG. 3 depicts a block diagram of a system for accessing data in accordance with an illustrative embodiment;

(5) FIG. 4 depicts a block diagram for a process to communicate data and control access in the system for managing and coordinating data in accordance with an illustrative embodiment;

(6) FIG. 5 each depict a block diagram showing example embodiments of a system for managing and coordinating data in accordance with an illustrative embodiment;

(7) FIGS. 6A-B each depict a block diagram showing example embodiments of a system for managing and coordinating data in accordance with an illustrative embodiment; and

(8) FIG. 7 is a block diagram of a server system and a client computer system in accordance with an illustrative embodiment.

DETAILED DESCRIPTION

(9) For purposes of reading the description of the various embodiments below, the following enumeration of the sections of the specification and their respective contents may be helpful: Section A describes systems and methods for managing and coordinate data for digital therapeutics trials; and Section B describes a network and computing environment, which may be useful for practicing embodiments described herein.

A. Systems and Methods for Managing and Coordinating Data for Digital Therapeutics Trials

(10) Referring now to FIG. 1, depicted is a block diagram of a system **100** for managing and coordinating data. In an overview, the system **100** may include at least one data management service **105**, at least one user device **130**, and a set of participant devices **110A-N** (hereinafter generally referred to as participant devices **110**), communicatively coupled with one another via at least one network **115**. At least one participant device **110** (e.g., the first participant device **110A** as depicted) may include at least one digital therapeutics application **120**. The user device **130** may include at least one interface application **140** (sometimes herein referred to as a data management application or portal application). The data management service **105** may include at least one data manager **155** and at least one access handler **160**, among others. The data management service **105** may include or have access to a data storage **175**. The data storage **175** may store, maintain, or otherwise include one or more datasets **180A-N** (hereinafter generally referred to as datasets **180**) among others. The data storage **175** may form or may be part of a data lake **185**. In some embodiments, the functionality of the digital therapeutics application **120** and the functionality of the interface application **140** may be performed in part on the data management service **105**.

(11) In further detail, the data management service **105** may (sometimes herein generally referred to as a computing system or a service) be any computing device comprising one or more processors coupled with memory and software and capable of performing the various processes and tasks described herein. The data management service **105** may be in communication with the one or more participant devices **110** and the data storage **175** via the network **115**. The data management service **105** may be situated, located, or otherwise associated with at least one server group. The server group may correspond to a data center, a branch office, or a site at which one or more servers corresponding to the data management service **105** is situated.

(12) Within the data management service **105**, the data manager **155** may manage data associated with data that is stored or to be stored in the data storage **175**. The data manager **155** may execute, initiate, or receive data from the digital therapeutics application **120** or the participant device **110**. The data manager **155** may execute, initiate, or receive data from the interface application **140** or the user device **130**. The data manager **155** may communicate the data stored in the data storage **175** to any of the participant device **110** or the user device **130**. For example, the data manager **155** may receive data from the participant device **110A**, and may store the data in the data storage **175**. In addition, the data manager **155** may retrieve data from the data storage **175**, and may send the data to the user device **130**. In some embodiments, the data manager **155** may send/receive data based at least on an indication from the access handler **160**.

(13) Within the data management service **105**, the access handler **160** may control access to data that is stored or to be stored in the data storage **175**. The access handler **160** may identify or verify the user device **130** to determine whether to provide access to data to the user device **130**. Upon a determination that the user device **130** may be granted the access, the access handler **160** can provide the access to the user device **130**. Upon a determination that the user device **130** should not be granted the access, the access handler **160** can deny a request for the access. For example, the access handler **160** can identify or verify the user device **130** based on identification information of the user device **130**.

(14) The participant device **110** may be any computing device comprising one or more processors coupled with memory and software and capable of performing the various processes and tasks described herein. The participant device **110** may be operated by a participant (sometimes herein referred to as a subject, patient, or user). The participant device **110** may be in communication with the data management service **105** and the data storage **175** via the network **115**. The participant device **110** may be a smartphone, other mobile phone, tablet computer, wearable computing device (e.g., smart watch, eyeglasses), or laptop computer. The participant device **110** may be used to access the digital therapeutics application **120**. In some embodiments, the digital therapeutics application **120** may be downloaded and installed on the participant device **110** (e.g., via a digital distribution platform). In some embodiments, the digital therapeutics application **120** may be a web application with resources accessible via the network **115**.

(15) The digital therapeutics application **120** executing on the participant device **110** may provide a digital therapeutics, and may facilitate a session (sometimes referred to herein as a therapy session) to address at least one condition of the user. The condition of the user may include, for example, a chronic pain (e.g., associated with or include arthritis, migraine, fibromyalgia, back pain, Lyme disease, endometriosis, repetitive stress injuries, irritable bowel syndrome, inflammatory bowel disease, and cancer pain), a skin pathology (e.g., atopic dermatitis, psoriasis, dermatillomania, and eczema), a cognitive impairment (e.g., mild cognitive impairment (MCI), Alzheimer's, multiple sclerosis, and schizophrenia), and other ailments (e.g., narcolepsy and oncology), among others.

(16) The user may be at least partially concurrently taking a pharmaceutical (sometimes referred herein as a medication or drug) to address the condition, while being provided sessions through digital therapeutics application **120**. For instance, if the medication is for pain, the user may be taking acetaminophen; a nonsteroidal anti-inflammatory composition; an antidepressant, an anticonvulsant; or other composition, among others. For skin pathologies, the user may be taking a

steroid, antihistamine, or topical antiseptic, among others. For cognitive impairments, the user may be taking cholinesterase inhibitors or memantine, among others. For narcolepsy, the user may be taking a stimulant or antidepressant, among others. The user of the digital therapeutics application **120** may also participate in other psychotherapies for these conditions.

(17) The digital therapeutics application **120** can include, present, or otherwise provide a user interface including one or more UI elements to a participant via the participant device **110** in accordance with a configuration on the digital therapeutics application **120**. The UI elements may correspond to visual components of the user interface, such as a command button, a text box, a check box, a radio button, a menu item, and a slider, among others. In some embodiments, the digital therapeutics application **120** may provide a session (sometimes referred to herein as a therapy session) via the user interface to achieve a behavioral endpoint of the participant (sometimes herein referred to as a patient, person, user, or subject). A behavioral endpoint can be, for example, a completion of the session, a physical or mental goal of a participant, a completion of a medication regimen, or a behavioral endpoint indicated by a doctor. A participant may be an individual who participates in the clinical trial. A participant may be enrolled in and participating in a digital therapeutics trial using the digital therapeutics application **120** on the participant device **110**.

(18) The user device **130** (sometimes herein referred to as an end user computing device) may be any computing device comprising one or more processors coupled with memory and software and capable of performing the various processes and tasks described herein. The user device **130** may be in communication with the data management service **105** and the data storage **175** via the network **115**. The user device **130** may be a smartphone, other mobile phone, tablet computer, wearable computing device (e.g., smart watch, eyeglasses), or laptop computer. The user device **130** may be used to run the interface application **140** to interface with the data management service **105** to access data on the data storage **175**. In some embodiments, the interface application **140** may be downloaded and installed on the user device **130** (e.g., via a digital distribution platform). In some embodiments, the interface application **140** may be a web application with resources accessible via the network **115**.

(19) A number of different types of users may operate the interface application **140** executing on the user device **130**. A user may include any individual who accesses the data management service **105** and can be, for example, a trial coordinator, data manager, maintenance engineer/programmer, or other sponsor representative. A sponsor may be a research sponsor, developer of a digital therapeutics, backer of the research, or owner or lead responsible for research conduct and may be responsible for trial management, trial design, and regulatory interactions. A sponsor may access the data management service **105** for creating and managing the trial(s) and evaluating the endpoint data.

(20) In addition, there may be other entities or individuals associated with the trials and entry of data via the interface application **140**. For instance, a clinical trial coordinator may be an individual who executes the research trial and collects data for entry through the data management service **105** via the interface application **140**. The clinical trial coordinator may interact directly with research participants. A trial coordinator may access the data management service **105** for approving or rejecting a participant's enrollment and selecting endpoint thresholds or other trial criteria. A principal investigator may be an individual who is responsible for compliant execution of the trial; attributable, legible, contemporaneous, original, and accurate (ALCOA) data collection; and Good Clinical Practice (GCP) compliance. The principal investigator may interact directly with research participants. A principal investigator may access the data management service **105** for assessing compliance to enrollment criteria and compliance with data security of PII and PHI.

(21) Furthermore, site staff may refer to the principal investigator, clinical trial coordinator, or other individuals responsible for research participant management. Site staff may access the data management service **105** to use participant contact information for participant ID verification and

enrollment. This identifying information is used to verify the identity of the participant and is accessible only to site staff responsible for the trial administration and participant interaction and communication. Records of participant informed consent are included in this information. The clinical trial database manager may be responsible for building and validating a clinical trial database. The data manager may also be responsible for addressing issues and issuing queries to principal investigators if necessary. The data manager may access the data management service **105** for defining access control mechanisms for data, for instance, by assigning roles to individuals, where the role corresponds to an access level for various data types.

(22) The data storage **175** may store and maintain various resources and datasets **180** associated with the data management service **105** and the digital therapeutics application **120**. In some embodiments, the data storage **175** may be part of, may form, or may otherwise be the data lake **185**. The data lake **185** may be a repository to store and maintain the data in both structured and unstructured formats. The data lake **185** may be used to store one or more data storages **175** storing datasets **180** from one or more trials. The data lake **185** may correspond to or include one or more storages (e.g., distributed system or cloud storage) for maintaining data in an unstructured or structured format. In some embodiments, the data storage **175** may include a database management system (DBMS) to arrange and organize at least a portion of the data maintained thereon. The data storage **175** may be in communication with the data management service **105** and the one or more participant devices **110** via the network **115**. While running various operations, the data management service **105** and the digital therapeutics application **120** may access the data storage **175** to retrieve identified data therefrom. The data management service **105** and the digital therapeutics application **120** may also write data onto the data storage **175** from running such operations.

(23) The datasets **180** stored and maintained on the data storage **175** may be aggregated from participants (e.g., directly or indirectly through a clinical trial site staff) and from the digital therapeutics application **120**. One type of datasets **180** may include digital therapeutics user data, such as data about a user's interaction with activities or a user's engagement with the digital therapeutics. This may be analogous to sites in pharmaceutical clinical trials monitoring adherence to the pharmaceutical. Another type of datasets **180** collected may include Electronic Patient Reported Outcome (ePRO). ePRO may include clinical outcome assessments self-reported by participants, generally via a validated questionnaire.

(24) Continuing on, another type of datasets **180** may include trial data that is entered by the participant (e.g., demographic data, eligibility questionnaires, etc.). This information may include validated clinical scales (C-SSRS, PHQ-9, etc.) or created forms and questionnaires (demographics, etc.). Another type of datasets **180** collected may include application metadata or application usage data collected automatically from the digital therapeutics data **120**. Specific data collected may be defined and configured by a data manager using tools or functions in a clinical trial facilitation framework.

(25) Referring now to FIG. 2, depicted is a block diagram of a process **200** for managing data in the system **100** for managing and coordinating data. The process **200** may correspond to or include operations in the system **100** to manage datasets **180** on the data storage **175**. Under the process **200**, the interface application **140** executing on the user device **130** may transmit or send at least one trial setup request **205** to the data management service **105**. The trial setup request **205** may include a set of parameters to define one or more trials for digital therapeutics. The set of parameters may be defined by a user (e.g., clinical trial coordinator, site staff, or clinical trial manager) of the interface application **140**. The set of parameters of the trial setup request **205** may identify or include digital therapeutics information (e.g., an identifier for the digital therapeutics application **120**), a set of participants **210** to use the digital therapeutics (e.g., anonymized identifiers or segment identifier), one or more phases **230A-N** (hereinafter generally referred to as phases **230**) (e.g., pre-trial phase, clinical trial phase, and post-trial phase), and a respective

endpoint (e.g., behavioral, psychological, or mental) for each phase via the digital therapeutics, among others.

(26) In some embodiments, the parameters of the trial setup request **205** may identify or include a set of questionnaires to provide the participants **210** for at least one phase **230** of the trial of the digital therapeutic. For example, the trial setup request **205** may include an upfront questionnaire, a lesson, instruction, biomarkers to be measured, a post-trial questionnaire, or otherwise any information for setting up a trial or for determining a status of the participant during the trial. In some embodiments, the trial setup request **205** may identify users (e.g., a clinical trial site staff) of the interface application **140** and a list of access privileges for each user for data associated with the trial. The access privilege may specify which type of data each user of the interface application **140** can access (e.g., read, write, or edit).

(27) The data manager **155** executing on the data management service **105** may receive, retrieve, or otherwise identify the trial setup request **205** from the user device **130**. The data manager **155** may parse the trial setup request **205** to identify information associated with trials for digital therapeutics. From parsing, the data manager **155** may identify the digital therapeutics application **120** to be tested for efficacy, safety, and performance on the participants **210** for the phases **230** of the trial. In some embodiments, the data manager **155** may identify a set of digital therapeutics applications **120** from the trial setup request **205**. The set of digital therapeutics applications **120** may be for different conditions (e.g., chronic pain, skin pathology, or multiple sclerosis). In some embodiments, the set of the digital therapeutics applications **120** may be for a single type of application used for different conditions or several different types of applications for the respective conditions.

(28) In some embodiments, the data manager **155** may identify a set of participants **210** to be included in the trial from the trial setup request **205**. The set of participants **210** may include users of the digital therapeutics applications **120**. Each of the participants **210** may be identified using an anonymized identifier. For example, the identifier may correspond to an account identifier for logging into the digital therapeutics application **120** or a randomly generated name used by a clinical trial site staff to uniquely reference the participant **210**. In some embodiments, the data manager **155** may identify the users (e.g., a clinical trial site staff) of the interface application **140** and the list of access privileges for each user from the digital therapeutics application **140**. In some embodiments, the data manager **155** may identify the set of questionnaires to provide the participants **210** for at least one phase **230** of the trial of the digital therapeutic treatment.

(29) In some embodiments, the data manager **155** may identify a set of trials from the trial setup request **205**. Each trial may define or include a set of phases **230** (sometimes herein referred to as stages), such as a pre-trial phase, a clinical trial phase to test the digital therapeutics application **120**, and a post-trial phase, among others. The set of trials may be for different conditions of participants **210** using the digital therapeutics application **120** to be tested for at least one of efficacy, safety, or performance on the participants **210**. The set of trials for different conditions may include, for example, chronic pain, a skin pathology, a cognitive impairment, and other ailments, among others. The set of trials for different conditions may include one or more pharmaceutical conditions. The set of trials for different conditions may be for a set of groups with different severity of a single condition.

(30) With the identifications, the data manager **155** may initialize, configure, and establish one data storage **175** in accordance with the parameters for the trial. The data storage **175** may be used to store data across phases **230** of a given trial for one or more digital therapeutics (e.g., one or more digital therapeutics applications **120**). For each trial, the data manager **155** may instantiate or establish a respective data storage **175** to store one or more datasets **180** for the trial. For instance, the data manager **155** may create a first data storage **175** for a trial testing the efficacy of a digital therapeutic and a second data storage **175** for a trial testing the safety of the digital therapeutics. Each trial may have three phases **230**, such as a pre-trial, a clinical trial, and a post-trial phase. The

data storage **175** may be established in or to form the data lake **185**. In some embodiments, the data storage **175** may correspond to the datasets **180** corresponding to a given trial within the data lake **185**. In some embodiments, the data storage **175** may correspond to or define a portion of the data lake **185** in which the datasets **180** for a given trial across one or more phases **230** is to be stored and maintained.

(31) With the establishment, the data manager **155** may manage, administer, or otherwise maintain the data storage **175** for the set of phases **230** (e.g., a pre-trial, clinical trial, and post-trial) associated with the trial of the digital therapeutics on the participants **210**. The data storage **175** may include a set of datasets **180**. The set of datasets **180** may be associated with one or more phases **230** of the trial for the digital therapeutics. At least one portion of datasets **180** may be from the participants **210** partaking in a corresponding phase **230** of the trial. For instance, the datasets **180** for the pre-trial phase **230A** or a post-trial phase **230C** may be acquired from answers to questionnaires by participants **210**. The questionnaires may be provided to the participants **210** directly via a prompt (e.g., on the digital therapeutics application **120** or another application) or via a clinician site staff (e.g., using the interface application **140** to enter data). At least one portion of datasets **180** may be from the digital therapeutics in a corresponding phase **230** of the trial. The data may be obtained or received from the digital therapeutics application **120** during the clinical trial phase **230B**, and may include, for example, metadata or log data identifying interactions by the participants **210** with the digital therapeutics application **120** running on the participant device **110**.

(32) In maintaining, the data manager **155** may retrieve, identify, or otherwise receive application data **220** from the digital therapeutics application **120**. The application data **220** may identify or include information generated by the digital therapeutics application **120** running on the participant device **110**. For example, the application data **220** may include: log data identifying one or more interactions by the participant **210** with the digital therapeutics application **120** or events triggered by processes while running the digital therapeutics application **120**; a completion level for the participant **210** through the lessons provided by the digital therapeutics application **120**; a performance score of the participant **210** in carrying out tasks as directed through the lessons; physiological measurements of the participant **210** while using the digital therapeutics application **120**; and metadata associated with the digital therapeutics application **120** (e.g., version identifier, network address, geographic data, device identifier, anonymized identifier, and target condition to be addressed) or identifying which phase **230** the application data **220** is obtained, among others. The application data **220** may be retrieved during at least one of the phases **230** (e.g., the clinical trial phase **230B**) of the trial.

(33) In conjunction, the data manager **155** may retrieve, identify, or otherwise receive participant data **225** from the participant **210**. The participant data **225** may be received via the digital therapeutics application **120**, the interface application **140**, or another application operated by clinical trial site staff, among others. In some embodiments, the participant data **225** may be sent to the data management service **105** without using the participant device **110**. For example, the participant **210** may directly communicate with a doctor or a clinician, who then may input or request to store the communicated information in the data management service **105** or in the data storage **175**.

(34) The participant data **225** may include or identify information about the participant **210** acquired outside the digital therapeutics application **120**. For example, the participant data **225** may identify or include one or more responses to a questionnaire; one or more traits of the participant **210** (e.g., age, race, gender, geographic location, or demographics); a condition of the participant **210** to be addressed via the digital therapeutics application **120**; a severity of the condition in the participant **210**; various physiological measurements taken from the participant **210** by a clinical trial site staff; diagnosis or evaluation information inputted by a clinician (e.g., a doctor or nurse) regarding the participant **210**; data directly from the participant **210** (e.g., electronic patient-reported outcomes (ePRO)); and metadata identifying which phase **230** the participant data **225** is

obtained, among others. In some embodiments, the participant data **225** may be generated in accordance with electronic data capture (EDC), new identity verification for the participant **210**, and informed consent on the part of the participant **210**. The participant data **225** may be retrieved during at least one of the phases **230** (e.g., the pre-trial phase **230A** or post-trial phase **230C**) of the trial.

(35) With the receipt of the new data (e.g., the application data **220** or participant data **225**), the data manager **155** may create, write, or otherwise generate at least one new dataset **180'** to store onto the data storage **175**. Using the new data, the data manager **155** may generate the new dataset **180'** to include the information included therein. The data manager **155** may parse the newly received data to identify which phase **230** in the trial the data is associated with. With the identification of the phase **239**, the data manager **155** may store or include the new dataset **180'** with the portion of the datasets **180** corresponding to the phase **230**. For example, the data manager **155** may update the portion of the datasets **180** corresponding to the first phase **230A** when the newly received data (e.g., the application data **220** or the participant data **225**) is identified as associated with the first phase **230A**.

(36) In some embodiments, the data manager **155** may generate the dataset **180'** or configure at least a portion of the data storage **175**, using electronic data capture, identity verification, and informed consent for participant data **225**. The data manager **155** may generate a shell or framework that can be used for various types of trials, phases, or digital therapeutics applications. In some embodiments, regardless of endpoints for each trial and procedures involved running the trials, the data manager **155** may provide a single framework for housing multiple trials. In some embodiments, the data manager **155** or created shells or frameworks are configured to capture specific clinical data from the participant devices **110** (e.g., ePRO data, app metadata, authentication data records, audit trail records, participant identification, and informed consent, etc.). The captured data may be provided to the user device **130** (e.g., the response **310**, as discussed below in greater detail).

(37) In conjunction, the data manager **155** may de-duplicate the datasets **180** across the phases **230** on the data storage **175**. The data manager **155** may identify the duplicated information in the dataset **180** in one portion (e.g., associated with the phase **230A**) and the dataset **180** in another portion (e.g., associated with the phase **230B**). With the identification, the data manager **155** may remove the duplicated information from at least one of the identified datasets **180**. For example, when the data manager **155** identifies both the datasets **180A** and the datasets **180B** include duplicated data associated with the phases **230A** and **230B** respectively, the data manager **155** may determine to remove the duplicated data from the datasets **180B**. In this example, both the datasets **180A** and **180B** may identify demographic information for the same participant **210**, and the data manager **155** may reduce the instances of the same demographic information by removing the data from one of the datasets **180A** or **180B**.

(38) In some embodiments, the data manager **155** may associate or link multiple data storages **175** from multiple trials to provide the interface application **140** access to the linked set of data storages **175** (or portions of datasets **180** across different data storages **175**). The data management service **105** may provide for plug-and-play functionality of the data storages **175** from multiple trials. For instance, the plug-and-play functionality may allow linkage or association of a first data storage **175** containing datasets **180** for a trial to test a safety of a digital therapeutic, with a second storage **175** containing datasets **180** for a trial to test an efficacy of the digital therapeutic. In some embodiments, the data manager **155** may retrieve, identify, or receive a request to link one data storage **175** from one trial with another data storage **175** from another trial from the interface application **140**. The request to link may be a part of the trial setup request **205**, and may identify the data storages **175** corresponding to different trials to be linked. In some embodiments, the request may identify the portions of datasets **180** from the different trials stored on corresponding data storages **175** to be linked.

(39) Upon receipt, the data manager **155** may determine, select, or otherwise identify the data storages **175** from multiple trials to be linked as identified in the request. In some embodiments, the data manager **155** may identify portions of datasets **180** across multiple data storages **175** from multiple trials to be linked as defined in the request. With the identification, the data manager **155** may link or associate the data storages **175** from multiple trials. In some embodiments, the data manager **155** may link the portions of datasets **180** across multiple data storages **175** from multiple trials. The data manager **155** may aggregate, join, or otherwise combine the data storages **175** from multiple trials in order to link them. The data manager **155** may assign the portions of datasets **180** identified in the requests to the data storages **180** from multiple trials of one another. To assign, the data manager **155** may generate an identifier to indicate that the portions of dataset **180** are associated with the multiple data storages **175** from multiple trials. With the linking of the multiple data storages **175** from multiple trials, the data manager **155** may enable or provide the user of the interface application **140** access to the linked data storages **175** from multiple trials with a single query or request.

(40) Referring now to FIG. **3**, depicted is a block diagram of a process **300** for accessing data in the system **100**. The process **300** may correspond to or include operations in the system **100** to access data on the data storage **175**. Under process **300**, the interface application **140** on the user device **130** may transmit or send an access request **305** to the data management service **105**. The access request **305** may be to query for datasets **180** stored on the data storage **175** for viewing or editing, or to store new data onto the data storage **175**. The access request **305** may include or identify one or more constraints for identifying one or more portions of the datasets **180** stored on the data storage **175** to be accessed.

(41) In some embodiments, the access request **305** may include or identify the phase **230** of the trial for which the datasets **180** on the data storage **175** are to be accessed. For example, the access request **305** may identify a pre-trial phase **230A**, a clinical phase **230B**, or a post-trial phase **230C**, for which associated datasets **180** are to be accessed. In some embodiments, the access request **305** may include or identify one or more participants **210** (e.g., using anonymized identifiers, traits, or other characteristics) for which the datasets **180** are to be accessed. For instance, the access request **305** may identify a particular participant **210** or a cohort of participants **210**, for which associated datasets **180** are to be accessed. In some embodiments, the access request **305** may include or identify a particular digital therapeutic application **120** for which datasets **180** are to be accessed. In some embodiments, the access request **305** may include or identify a particular condition for which datasets **180** are to be accessed. The requested datasets **180** may be stored or maintained on one data storage **175** that is a data lake.

(42) The access request **305** may identify or include other information in connection with accessing the datasets **180**. The access request **305** may also identify a type of operation to be performed on the datasets **180**, such as read, write, view, or create, among others. In some embodiments, the access request **305** may include new data to be stored or entered onto the data storage **175**, similar to the application data **220**, participant data **225**, or trial setup request **205** (e.g., at least one questionnaire). For example, the access request **305** may include a list of questionnaires to measure the safety, efficacy, or performance of the digital therapeutics application **120** to be provided to the participants of a given trial. In some embodiments, the access request **305** may include or identify information indicating an identification of a clinician, clinic, or user device based on which the access handler **160** may determine whether to provide the access. For example, the access request **305** may include a registration number of a clinician, clinic, or user device that requests the access.

(43) The access handler **160** executing on the data management service **105** may retrieve, identify, or otherwise receive the access request **305** from the interface application **140** on the user device **130**. Upon receipt, the access handler **160** may parse the access request **305** to extract or identify the one or more constraints identified therein for finding a corresponding portion of dataset **180**. Based on the constraint, the access handler **160** may identify or select a corresponding portion of

the datasets **180''** on the data storage **175**. In some embodiments, the access handler **160** may identify or select the portion of datasets **180''** corresponding to the phase **230** identified in the access request **305**. In some embodiments, the access handler **160** may select the portion of datasets **180''** corresponding to the digital therapeutics application **120** identified in the access request **305**. In some embodiments, the access handler **160** may select the portion of datasets **180''** corresponding to the particular condition tested in the trial identified in the access request **305**.

(44) In some embodiments, the access handler **160** may select the portion of datasets **180''** corresponding to the participants identified in the access request **305**. The access handler **160** may associate, map, or otherwise correlate datasets **180** from the different portions (or phases **230**) for the identified participant **210**. For example, the access handler **160** may identify the datasets **180** in each portion (or phase **230**) associated with the participant **210**. The datasets **180** may be taken from the application data **220** generated when the participant **210** used the digital therapeutics application **120**. The datasets **180** may also be obtained from the participant data **225** obtained via questionnaire presented by a clinical trial site staff. With the identifications, the access handler **160** may correlate the datasets **180** as associated with the same participant **210**.

(45) In some embodiments, the access handler **160** may determine whether to grant or permit access to the user device **130** to the selected portion of the datasets **180''** in the data storage **175** in accordance with access privileges for the user. The access privileges may be defined for the user of the device **130** in the trial setup request **205** used to establish the data storage **175**, and may specify which types of operations the user is permitted (e.g., read, write, view, or create). The access handler **160** may identify the access privilege for the user as defined in establishment of the data storage **175**. With the identification, the access handler **160** may determine whether the user device **130** is to be permitted access to the portion of the datasets **180''**. The determination may be on a per dataset **180''** basis. When the access privilege permits access, the access handler **160** may determine to grant the user device **130** access to the selected portion of the datasets **180''** for the permitted types of operations. Otherwise, when the access privilege denies access, the access handler **160** may determine to restrict or deny the user device **130** access to the selected portion of the datasets **180''** for the unpermitted types of operations.

(46) In some embodiments, the access handler **160** may control or regulate access to datasets **180''** based on authentication information from the user device **130**. The access handler **160** may identify authentication information from the access request **305**. For example, the authentication information may include a user identifier and password for the user of the user device **130**, among others. In some embodiments, the access handler **160** may verify the user device **130** based on the credentials in the access request **305**. If the verification is successful, the access handler **160** may determine to grant the user device **130** access to the selected portion of the datasets **180''** for the permitted types of operations. Otherwise, if the verification is a failure, the access handler **160** may determine to restrict or deny the user device **130** access to the selected portion of the datasets **180''** for the unpermitted types of operations.

(47) With the selection, the access handler **160** may provide the user device **130** access to the portion of the datasets **180''**. In some embodiments, the provision of the access may be in response to determining to grant based on access privileges or in response to successful verifications of the authentication credentials. The access handler **160** may interface with the interface application **140** for the access. In some embodiments, the access handler **160** may transmit, provide, or otherwise send at least one response **310** to the user device **130**. The response **310** may identify or include the portion of the datasets **180''** to provide to the interface application **140** on the user device **130**. In some embodiments, the access handler **160** may generate a new dataset **180** to include onto the data storage **175** when the access request **305** is to store new data. For instance, the new dataset **180** may include list of questionnaires to measure the safety, efficacy, or performance of the digital therapeutics application **120** to be provided to the participants of a given trial, as defined in the access request **305**. The new dataset **180** may be included or identified in the response **310**.

(48) With the access, the interface application **140** on the user device **130** may generate an output for presentation using the datasets **180** identified in the response **310**. The user device **130** may include a display to present, display, list, or otherwise show at least a portion of the datasets **180** on the display. The interface application **140** can control or manage displaying at least a portion of the datasets **180** A-N upon receipt of the same. In some embodiments, the interface application **140** may generate information based on the datasets **180**. For example, the interface application **140** may present information identifying a plurality of measurements for the efficacy of the digital therapeutics application across a plurality of phases. The information may indicate whether a percentage of participants **210** reaching endpoints (e.g., behavioral, psychological, or mental) in relation to the condition to be addressed by the digital therapeutics application **140**. In some embodiments, the interface application **140** may receive real-time data from the participant device **110**.

(49) In this manner, the data management service **105** together with the interface application **140** on the user device **130** may allow the user to participate in more than one digital therapeutics trial within the same infrastructure on their user device **130**. This may offer the advantage of providing the user with a single and consistent user interface for multiple trials. The user may have local data that is shared among the various trials because they are integrated into a single data storage **175**. For instance, the user may not need to re-enter personally identifiable information or personal health information multiple times for each phase **230** because the information is saved locally only once. Relative to approaches entailing the use of multiple, disparate databases to store data acquired from different phases and trials, the data management service **105** may reduce consumption of processing resources as well as space for data storage. The ability to access the datasets **180** on data storages **175** from one or more trials via the interface application **140** may also provide for improved quality of human-computer interaction (HCI) between the user and the other components of the system **100**.

(50) Referring now to FIG. 4, depicted is a flow diagram for a method **400** of managing and coordinating data. The method **400** may be performed by, but not limited to, any systems, devices, or components thereof described above with respect to FIGS. 1-3 or FIG. 7 below. At step **405**, a clinician user device (e.g., the user device **130**) sends a setup request (e.g., a trial setup request **205**) to a data management service (e.g., a data management service **105**). At step **410**, the data management service establishes, creates, configures, or generates a data storage or dataset responsive to the setup request, or setup the data storage or dataset for trials of digital therapeutic applications. At step **415**, a participant device (e.g., a participant device **110**) can send data (e.g., application data **220**, participant data **225**) to the data management service. At step **420**, the data management service can receive the data from the participant or the participant device. In response to receipt of the data, the data management service can store the data in the data storage. At step **425**, the user device can send a request for access (e.g., access request **305**) to the data to the data management service. At step **430**, the data management service can identify the user device based at least in part on the request for access. At step **435**, the data management service can provide the user device with access to the data (e.g., a response **310**). At step **440**, the user device may access the data to read, write, or edit.

(51) Referring now to FIG. 5, depicted is a block diagram of a system **500B** for managing and coordinating data. The data storage may form a data lake **505**. The data lake **505** may store a plurality of types of data, thereby integrating data for one or more trials of digital therapeutic applications. A first application **140A** may be used by study staff to access at least a portion (e.g., dataset **515A** for digital therapeutics application data, dataset **515B** for pre-trial data, or dataset **515C** for post-trial data) of the data lake **505** via a clinical portal. A second application **140B** may be used by a clinician manager to access at least a portion (e.g., the datasets **515A-C**) of the data lake **505** via, for example, a third party application (e.g., Snowflake). In some embodiments, the access of the first application **140A** and the second application **140B** may be determined based on,

for example, privilege information identified in the access request **305**.

(52) Referring now to FIG. **6A**, depicted is a block diagram **600A** showing example embodiments of a system for managing and coordinating data. In some embodiments, the digital therapeutics application **120** and the data lake **610** may include ePRO, which can communicate data between the digital therapeutics application **120** and a data lake **610**. The ePRO can capture or report clinical endpoints, pre-trial work, post-trial work, etc., from the digital therapeutics application **120**, and can send to the data lake **610**. In some embodiments, the digital therapeutics application **120** may include participant verification data (e.g., electronic consent, ID check, etc.) and can send to the data lake **610**. The data lake **610** can receive, identify, store, or control the data received from the digital therapeutics application **120**. For example, the data lake **610** may store an ePRO dataset, and participant verification dataset, as well as send at least portion of the same to applications **615**. The applications **615** may include a first application **615A**, and a second application **615B**. Each of the first application **615A** and the second application **615B** may play a different role. For example, the first application **615A** may be a front-end portal for the clinical team and may be used by a clinical trial coordinator.

(53) The first application **615A** may have access to or manage the ePRO in the digital therapeutics application **120**, while communicating data with the data lake **610**. The second application **615B** may be a part of the data lake **610**, including data tables for digital therapeutics data, and may be used by a clinical trial data manager or a clinician. For example, the second application **615B** may be connected to at least a portion of the data lake **610** (e.g., the ePRO, DTx dataset, etc.). In some embodiments, the second application **615B** may provide the first application **615A** with data for setting up, configuring, or establishing one or more datasets based at least in part on the ePRO dataset. In some embodiments, as shown in FIG. **6B**, the application **615B** may be excluded from the data lake **610**. The data lake **610** may include the ePRO, DTx dataset, and participant dataset, and can provide the applications **615**.

(54) B. Network and Computing Environment

(55) Various operations described herein can be implemented on computer systems. FIG. **7** shows a simplified block diagram of a representative server system **700**, client computer system **714**, and network **726** usable to implement certain embodiments of the present disclosure. In various embodiments, server system **700** or similar systems can implement services or servers, described herein, or portions thereof. Client computer system **714** or similar systems can implement clients described herein. The system **100** described herein can be similar to the server system **700**. Server system **700** can have a modular design that incorporates a number of modules **702** (e.g., blades in a blade server embodiment); while two modules **702** are shown, any number can be provided. Each module **702** can include processing unit(s) **704** and local storage **706**.

(56) Processing unit(s) **704** can include a single processor, which can have one or more cores, or multiple processors. In some embodiments, processing unit(s) **704** can include a general-purpose primary processor as well as one or more special-purpose co-processors such as graphics processors, digital signal processors, or the like. In some embodiments, some or all processing units **704** can be implemented using customized circuits, such as application specific integrated circuits (ASICs) or field programmable gate arrays (FPGAs). In some embodiments, such integrated circuits execute instructions that are stored on the circuit itself. In other embodiments, processing unit(s) **704** can execute instructions stored in local storage **706**. Any type of processors in any combination can be included in processing unit(s) **704**.

(57) Local storage **706** can include volatile storage media (e.g., DRAM, SRAM, SDRAM, or the like) and/or non-volatile storage media (e.g., magnetic or optical disk, flash memory, or the like). Storage media incorporated in local storage **706** can be fixed, removable, or upgradeable as desired. Local storage **706** can be physically or logically divided into various subunits such as a system memory, a read-only memory (ROM), and a permanent storage device. The system memory can be a read-and-write memory device or a volatile read-and-write memory, such as dynamic

random-access memory. The system memory can store some or all of the instructions and data that processing unit(s) **704** need at runtime. The ROM can store static data and instructions that are needed by processing unit(s) **704**. The permanent storage device can be a non-volatile read-and-write memory device that can store instructions and data even when module **702** is powered down. The term “storage medium” as used herein includes any medium in which data can be stored indefinitely (subject to overwriting, electrical disturbance, power loss, or the like) and does not include carrier waves and transitory electronic signals propagating wirelessly or over wired connections.

(58) In some embodiments, local storage **706** can store one or more software programs to be executed by processing unit(s) **704**, such as an operating system and/or programs implementing various server functions such as functions of the system **100** or any other system described herein, or any other server(s) associated with system **100** or any other system described herein.

(59) “Software” refers generally to sequences of instructions that, when executed by processing unit(s) **704**, cause server system **700** (or portions thereof) to perform various operations, thus defining one or more specific machine embodiments that execute and perform the operations of the software programs. The instructions can be stored as firmware residing in read-only memory and/or program code stored in non-volatile storage media that can be read into volatile working memory for execution by processing unit(s) **704**. Software can be implemented as a single program or a collection of separate programs or program modules that interact as desired. From local storage **706** (or non-local storage described below), processing unit(s) **704** can retrieve program instructions to execute and data to process in order to execute various operations described above.

(60) In some server systems **700**, multiple modules **702** can be interconnected via a bus or other interconnect **708**, forming a local area network that supports communication between modules **702** and other components of server system **700**. Interconnect **708** can be implemented using various technologies, including server racks, hubs, routers, etc.

(61) A wide area network (WAN) interface **710** can provide data communication capability between the local area network (e.g., through the interconnect **708**) and the network **726**, such as the Internet. Other technologies can be used to communicatively couple the server system with the network **726**, including wired (e.g., Ethernet, IEEE 802.3 standards) and/or wireless technologies (e.g., Wi-Fi, IEEE 802.11 standards).

(62) In some embodiments, local storage **706** is intended to provide working memory for processing unit(s) **704**, providing fast access to programs and/or data to be processed while reducing traffic on an interconnect **708**. Storage for larger quantities of data can be provided on the local area network by one or more mass storage subsystems **712** that can be connected to an interconnect **708**. Mass storage subsystem **712** can be based on magnetic, optical, semiconductor, or other data storage media. Direct attached storage, storage area networks, network-attached storage, and the like can be used. Any data stores or other collections of data described herein as being produced, consumed, or maintained by a service or server can be stored in a mass storage subsystem **712**. In some embodiments, additional data storage resources may be accessible via a WAN interface **710** (potentially with increased latency).

(63) Server system **700** can operate in response to requests received via a WAN interface **710**. For example, one of modules **702** can implement a supervisory function and assign discrete tasks to other modules **702** in response to received requests. Work allocation techniques can be used. As requests are processed, results can be returned to the requester via a WAN interface **710**. Such operation can generally be automated. Further, in some embodiments, a WAN interface **710** can connect multiple server systems **700** to each other, providing scalable systems capable of managing high volumes of activity. Other techniques for managing server systems and server farms (collections of server systems that cooperate) can be used, including dynamic resource allocation and reallocation.

(64) The server system **700** can interact with various user-owned or user-operated devices via a

wide-area network such as the Internet. An example of a user-operated device is shown in FIG. 7 as client computing system **714**. The client computing system **714** can be implemented, for example, as a consumer device such as a smartphone, other mobile phone, tablet computer, wearable computing device (e.g., smart watch, eyeglasses), desktop computer, laptop computer, and so on. (65) For example, the client computing system **714** can communicate via a WAN interface **710**. The client computing system **714** can include computer components such as the processing unit(s) **716**, storage device **718**, network interface **720**, user input device **722**, and user output device **724**. The client computing system **714** can be a computing device implemented in a variety of form factors, such as a desktop computer, laptop computer, tablet computer, smartphone, other mobile computing device, wearable computing device, or the like.

(66) The processing unit **716** and storage device **718** can be similar to processing unit(s) **704** and local storage **706** described above. Suitable devices can be selected based on the demands to be placed on client computing system **714**; for example, the client computing system **714** can be implemented as a “thin” client with limited processing capability or as a high-powered computing device. The client computing system **714** can be provisioned with program code executable by processing unit(s) **716** to enable various interactions with the server system **700**.

(67) Network interface **720** can provide a connection to the network **726**, such as a wide area network (e.g., the Internet) to which WAN interface **710** of server system **700** is also connected. In various embodiments, network interface **720** can include a wired interface (e.g., Ethernet) and/or a wireless interface implementing various RF data communication standards such as Wi-Fi, Bluetooth, or cellular data network standards (e.g., 3G, 4G, LTE, etc.).

(68) User input device **722** can include any device (or devices) via which a user can provide signals to client computing system **714**; client computing system **714** can interpret the signals as indicative of particular user requests or information. In various embodiments, user input device **722** can include any or all of a keyboard, touch pad, touch screen, mouse or other pointing device, scroll wheel, click wheel, dial, button, switch, keypad, microphone, and so on.

(69) User output device **724** can include any device via which client computing system **714** can provide information to a user. For example, user output device **724** can include display-to-display images generated by or delivered to client computing system **714**. The display can incorporate various image generation technologies, e.g., a liquid crystal display (LCD), light-emitting diode (LED) display including organic light-emitting diodes (OLED), projection system, cathode ray tube (CRT), or the like, together with supporting electronics (e.g., digital-to-analog or analog-to-digital converters, signal processors, or the like). Some embodiments can include a device such as a touchscreen that function as both input and output device. In some embodiments, other user output devices **724** can be provided in addition to or instead of a display. Examples include indicator lights, speakers, tactile “display” devices, printers, and so on.

(70) Some embodiments include electronic components, such as microprocessors, storage, and memory that store computer program instructions in a computer readable storage medium. Many of the features described in this specification can be implemented as processes that are specified as a set of program instructions encoded on a computer readable storage medium. When these program instructions are executed by one or more processing units, they cause the processing unit(s) to perform various operations indicated in the program instructions. Examples of program instructions or computer code include machine code, such as is produced by a compiler, and files including higher-level code that are executed by a computer, an electronic component, or a microprocessor using an interpreter. Through suitable programming, processing unit(s) **704** and **716** can provide various functionality for server system **700** and client computing system **714**, including any of the functionality described herein as being performed by a server or client, or other functionality.

(71) It will be appreciated that server system **700** and client computing system **714** are illustrative and that variations and modifications are possible. Computer systems used in connection with embodiments of the present disclosure can have other capabilities not specifically described here.

Further, while server system **700** and client computing system **714** are described with reference to particular blocks, it is to be understood that these blocks are defined for convenience of description and are not intended to imply a particular physical arrangement of component parts. For instance, different blocks can be but need not be located in the same facility, in the same server rack, or on the same motherboard. Further, the blocks need not correspond to physically distinct components. Blocks can be configured to perform various operations, e.g., by programming a processor or providing appropriate control circuitry, and various blocks might or might not be reconfigurable depending on how the initial configuration is obtained. Embodiments of the present disclosure can be realized in a variety of apparatus including electronic devices implemented using any combination of circuitry and software.

(72) While the disclosure has been described with respect to specific embodiments, one skilled in the art will recognize that numerous modifications are possible. Embodiments of the disclosure can be realized using a variety of computer systems and communication technologies, including but not limited to specific examples described herein. Embodiments of the present disclosure can be realized using any combination of dedicated components and/or programmable processors and/or other programmable devices. The various processes described herein can be implemented on the same processor or different processors in any combination. Where components are described as being configured to perform certain operations, such configuration can be accomplished, e.g., by designing electronic circuits to perform the operation, by programming programmable electronic circuits (such as microprocessors) to perform the operation, or any combination thereof. Further, while the embodiments described above may make reference to specific hardware and software components, those skilled in the art will appreciate that different combinations of hardware and/or software components may also be used and that particular operations described as being implemented in hardware might also be implemented in software or vice versa.

(73) Computer programs incorporating various features of the present disclosure may be encoded and stored on various computer readable storage media; suitable media include magnetic disk or tape, optical storage media such as compact disk (CD) or digital versatile disk (DVD), flash memory, and other non-transitory media. Computer readable media encoded with the program code may be packaged with a compatible electronic device, or the program code may be provided separately from electronic devices (e.g., via Internet download or as a separately packaged computer-readable storage medium).

(74) Thus, although the disclosure has been described with respect to specific embodiments, it will be appreciated that the disclosure is intended to cover all modifications and equivalents within the scope of the following claims.

Claims

1. A method of managing access to data associated with trials for digital therapeutics on participants, comprising: receiving, by a server, from a user device, a request to define a plurality of parameters for at least one trial, the plurality of parameters identifying at least one of a digital therapeutic, a plurality of phases, or a respective endpoint for each of the plurality of phases; establishing, by the server, a data lake comprising a data storage to store a plurality of datasets in at least one of a structured format or an unstructured format for the at least one trial in accordance with the plurality of parameters; maintaining, by the server, for multiple digital therapeutics, the data lake for a plurality of phases associated with a trial for a digital therapeutic provided by a digital therapeutic application to a plurality of participants, the plurality of phases including (a) a pre-trial phase prior to commencing the trial for the digital therapeutic, (b) a trial phase during the trial, and (c) a post-trial phase subsequent to completion of the trial for the digital therapeutic, wherein the data lake maintains the data of the plurality of phases in a centralized manner, the data lake comprising: (i) a plurality of datasets from the pre-trial phase, (ii) a plurality of datasets from

the trial phase, and (iii) a plurality of datasets from the post-trial phase; receiving, by the server, from the user device, an additional request associated with a participant or group of participants, an endpoint, and a phase of the plurality of phases for which the plurality of datasets is to be accessed; selecting, by the server, a corresponding portion of the plurality of datasets based on the additional request, wherein the corresponding portion is selected by identifying linked data linked by an association between a plurality of trials; and providing, by the server, to the user device, the corresponding portion of the plurality of datasets in the data lake.

2. The method of claim 1, further comprising: identifying, by the server, a plurality of trials associated with a corresponding plurality of conditions of the plurality of participants provided with the digital therapeutic; and establishing, by the server, a plurality of data storages for the corresponding plurality of trials.

3. The method of claim 1, further comprising: receiving, by the server, from the user device, a second request to associate the data storage with a second data storage, the second data storage comprising a second plurality of datasets from a second trial separate from the trial associated with the data storage; and linking, by the server, responsive to the second request, the second data storage with the data storage to provide the user device access to the data storage and the second data storage.

4. The method of claim 1, further comprising identifying, by the server, from a plurality of access privileges, an access privilege for a user associated with the user device, and wherein providing the corresponding portion further comprises providing the user device access to the portion of the plurality of datasets, responsive to the access privilege defining that the user is permitted to access the portion.

5. The method of claim 1, wherein receiving the request further comprises: receiving the request identifying a questionnaire to add to the phase of the plurality of phases; and generating, by the server, at least one dataset identifying the questionnaire to add to at least one corresponding portion of the plurality of datasets.

6. The method of claim 1, wherein maintaining the data storage further comprises removing, from the data storage, a first dataset in one portion of the plurality of datasets corresponding to a second dataset in another portion of the plurality of datasets.

7. The method of claim 1, wherein receiving the request further comprises receiving the request identifying a participant of the plurality of participants for which the plurality of datasets is to be accessed; wherein providing the access further comprises correlating, within the plurality of datasets on the data storage, one or more datasets associated with the participant identified in the request.

8. The method of claim 1, wherein at least one of the plurality of datasets are generated using at least one of electronic data capture (EDC), identity verification, and informed consent.

9. The method of claim 1, wherein at least one of the plurality of participants is administered with a pharmaceutical to address a condition, at least partially concurrently with use of the digital therapeutic.

10. A system for managing access to data associated with trials for digital therapeutics on participants, comprising: at least one server having one or more processors coupled with memory, configured to: receive, from a user device, a request to define a plurality of parameters for at least one trial, the plurality of parameters identifying at least one of a digital therapeutic, a plurality of phases, or a respective endpoint for each of the plurality of phases; establish a data lake comprising a data storage to store a plurality of datasets in at least one of a structured format or an unstructured format for the at least one trial in accordance with the plurality of parameters; maintain, for multiple digital therapeutics, the data lake for a plurality of phases associated with a trial for a digital therapeutic provided by a digital therapeutic application to a plurality of participants, the plurality of phases including (a) a pre-trial phase prior to commencing the trial for the digital therapeutic, (b) a trial phase during the trial, and (c) a post-trial phase subsequent to completion of

the trial for the digital therapeutic, wherein the data lake maintains the data of the plurality of phases in a centralized manner, the data lake comprising: (i) a plurality of datasets from the pre-trial phase, (ii) a plurality of datasets from the trial phase, and (iii) a plurality of datasets from the post-trial phase; receive, from the user device, an additional request associated with a participant or group of participants, an endpoint, and a phase of the plurality of phases for which the plurality of datasets is to be accessed; select a corresponding portion of the plurality of datasets based on the additional request, wherein the corresponding portion is selected by identifying linked data linked by an association between a plurality of trials; and provide, to the user device, the corresponding portion of the plurality of datasets in the data lake.

11. The system of claim 10, wherein the at least one server is further configured to: identify a plurality of trials associated with a corresponding plurality of conditions of the plurality of participants provided with the digital therapeutic; and establish a plurality of data storages for the corresponding plurality of trials.

12. The system of claim 10, wherein the at least one server is further configured to: receive, from the user device, a second request to associate the data storage with a second data storage, the second data storage comprising a second plurality of datasets from a second trial separate from the trial associated with the data storage; and link, responsive to the second request, the second data storage with the data storage to provide the user device access to the data storage and the second data storage.

13. The system of claim 10, wherein the at least one server is further configured to: identify, from a plurality of access privileges, an access privilege for a user associated with the user device; and wherein to provide the corresponding portion, the at least one server is configured to provide, to the user device associated with the user, access to the portion of the plurality of datasets, responsive to the access privilege defining that the user is permitted to access the portion.

14. The system of claim 10, wherein the at least one server is further configured to: receive the request identifying a questionnaire to add to the phase of the plurality of phases; and generate at least one dataset identifying the questionnaire to add to at least one corresponding portion of the plurality of datasets.

15. The system of claim 10, wherein the at least one server is further configured to remove, from the data storage, a first datasets in one portion of the plurality of datasets corresponding to a second dataset in another portion of the plurality of datasets.

16. The system of claim 10, wherein the at least one server is further configured to: receive the request identifying a participant of the plurality of participants for which the plurality of datasets is to be accessed; correlate, within the plurality of datasets on the data storage, one or more datasets associated with the participant identified in the request.

17. The system of claim 10, wherein at least one of the plurality of datasets are generated using at least one of electronic data capture (EDC), identity verification, and informed consent.

18. The system of claim 10, wherein at least one of the plurality of participants is administered with a pharmaceutical to address a condition, at least partially concurrently with use of the digital therapeutic.
